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Solving Inequalities

Toolbox

Solving Inequalities

Solving inequalities is much like solving equations. The goal is to use inverse operations to isolate the variable on one side of the inequality.

There is one difference from solving equations. If you multiply or divide both sides of an inequality by a negative number, you must change the direction of the inequality symbol.

Can you see why? This statement is true: $7 > 4$

Now multiply both sides by -1 . The statement is now false: $-7 > -4$

To make it true, flip the sign: $-7 > -4$ becomes $-7 < -4$.

* if we multiply or divide both sides by a negative, switch the inequality.

Example 1

Hide

Solve: $4x - 7 > 5$

Step 1

Add 7 to both sides.

$4x - 7 + 7 > 5 + 7$

$4x > 12$

Step 2

Divide both sides by 4.

$\frac{4x}{4} > \frac{12}{4}$

$x > 3$

The solution is $x > 3$.

$>$ greater than

$<$ less than

$\geq$ greater than or equal to

$\leq$ less than or equal to

$3 < 5$

big opening is next to the bigger value.

Example 2

Hide

Solve: $-9x + 8 < 26$

Step 1

Subtract 8 from both sides.

$-9x + 8 - 8 < 26 - 8$

$-9x < 18$

Step 2

Divide both sides by -9 AND flip the inequality symbol.

$\frac{-9x}{-9} > \frac{18}{-9}$

$x > -2$

The solution is $x > -2$.

Key Terms

- compound inequality
- inequality

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Category

Lesson 5: Inequalities

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